# NAND2TETRIS ASSIGNMENT

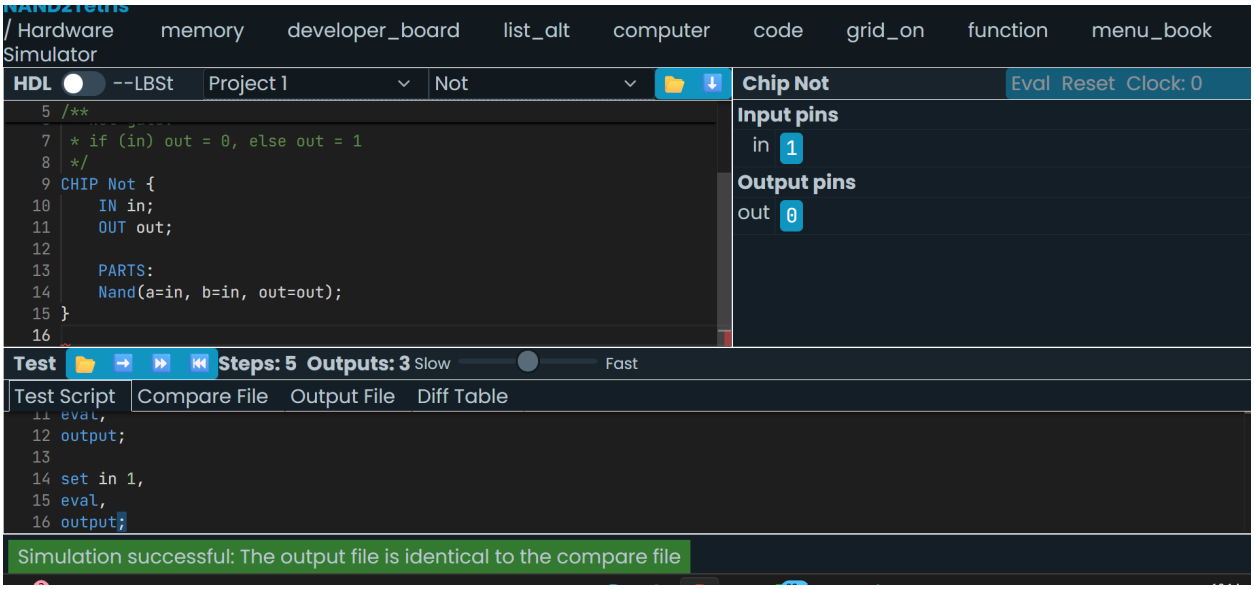
## Project 1

By - Tanvi Choudhary
Roll No: 2501420022

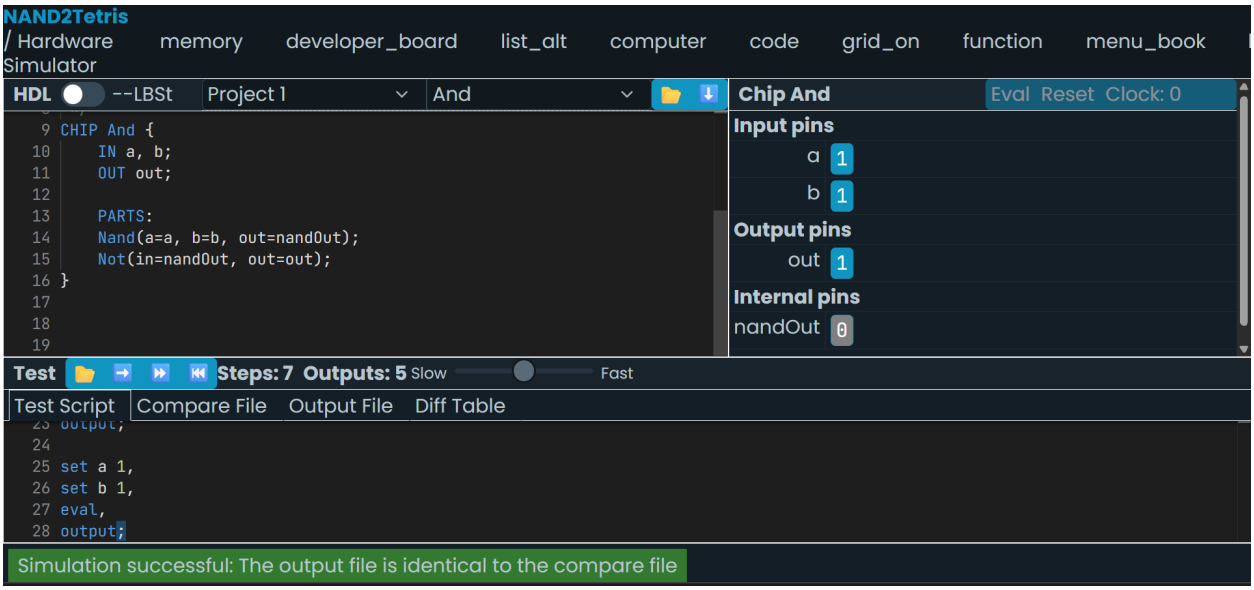
## Chips Implemented

1 Not
2 And
3 Or
4 Xor
5 Mux
6 DMux
7 Not16
8 And16
9 Or16
10 Mux16
11 Or8Way
12 Mux4Way16
13 Mux8Way16
14 DMux4Way
15 DMux8Way

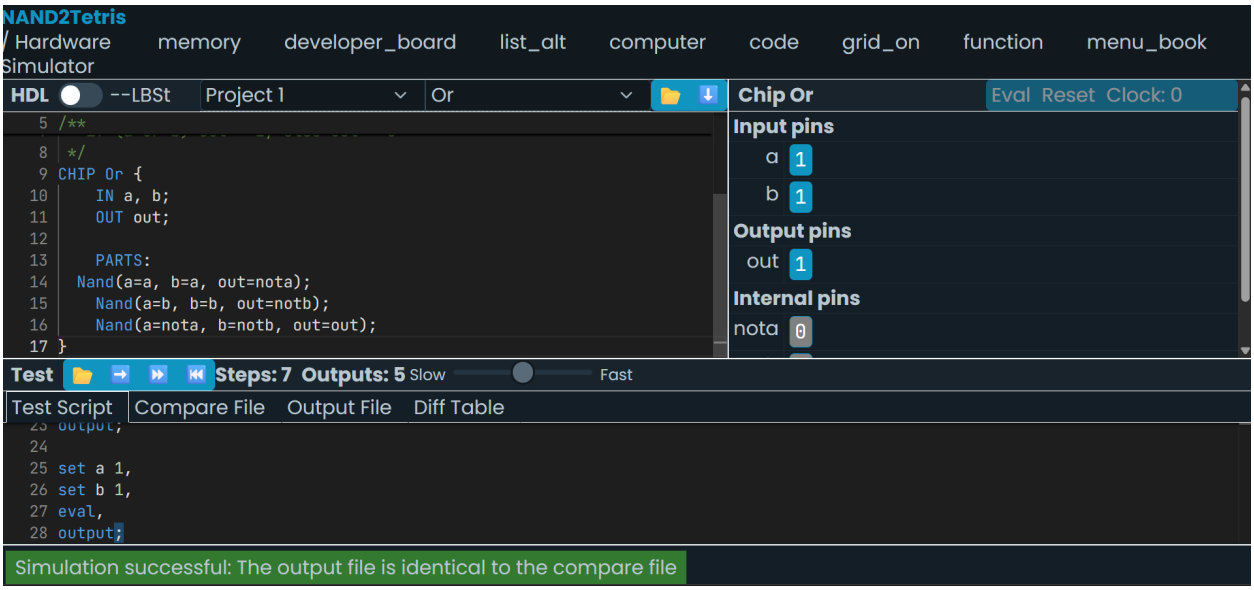
# Chip: Not


/ Hardware memory developer_board list_alt computer code grid_on function menu_book
Simulator
HDL --LBSt Project 1 Not
5 /**
7 * if (in) out = 0, else out = 1
8 */
9 CHIP Not {
10 IN in;
11 OUT out;
12
13 PARTS:
14 Nand(a=in, b=in, out=out);
15 }
16
Chip Not Eval Reset Clock: 0
Input pins
in 1
Output pins
out 0
Test Steps: 5 Outputs: 3 Slow Fast
Test Script Compare File Output File Diff Table
12 output;
13
14 set in 1,
15 eval,
16 output;
Simulation successful: The output file is identical to the compare file

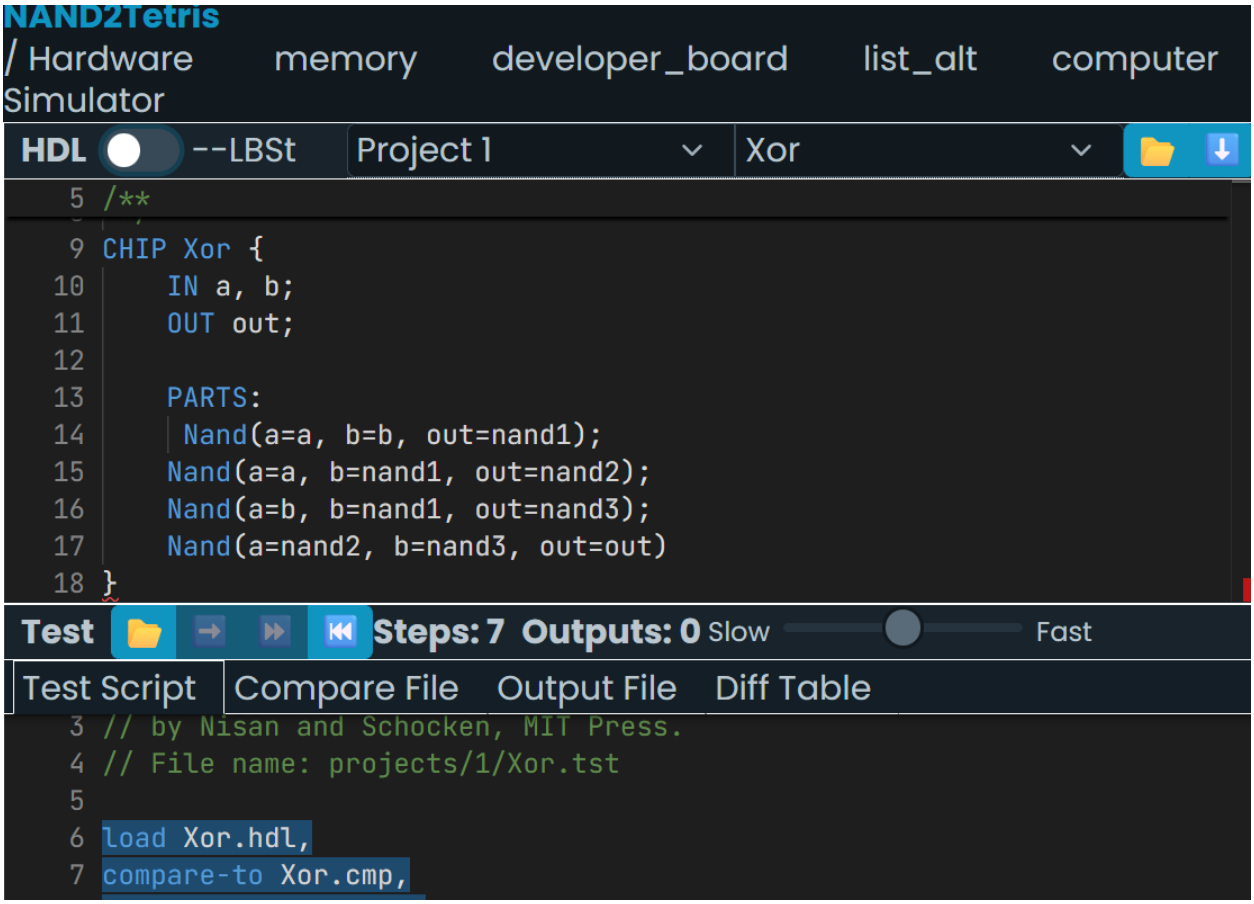
# Chip: And


NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
And
9 CHIP And {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a, b=b, out=nandOut);
15 Not(in=nandOut, out=out);
16 }
17
18
19
Chip And
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
nandOut 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

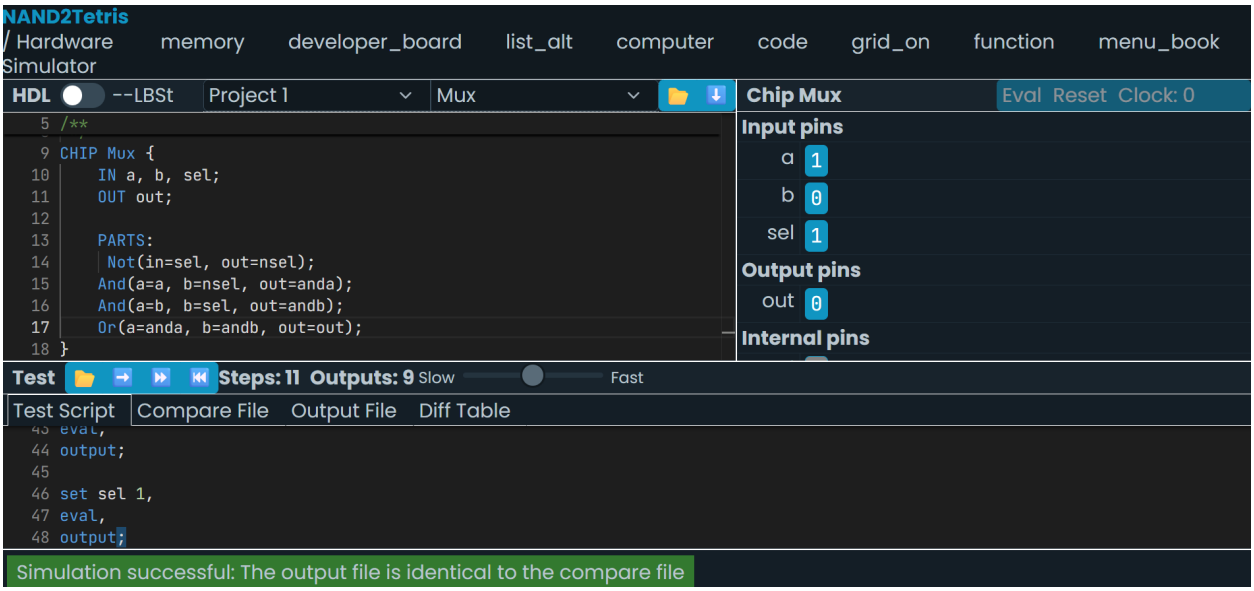
# Chip: Or


NAND2Tetris
Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
Or
5 /**
8 */
9 CHIP Or {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a, b=a, out=nota);
15 Nand(a=b, b=b, out=notb);
16 Nand(a=nota, b=notb, out=out);
17 }
Chip Or
Eval
Reset
Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
nota 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

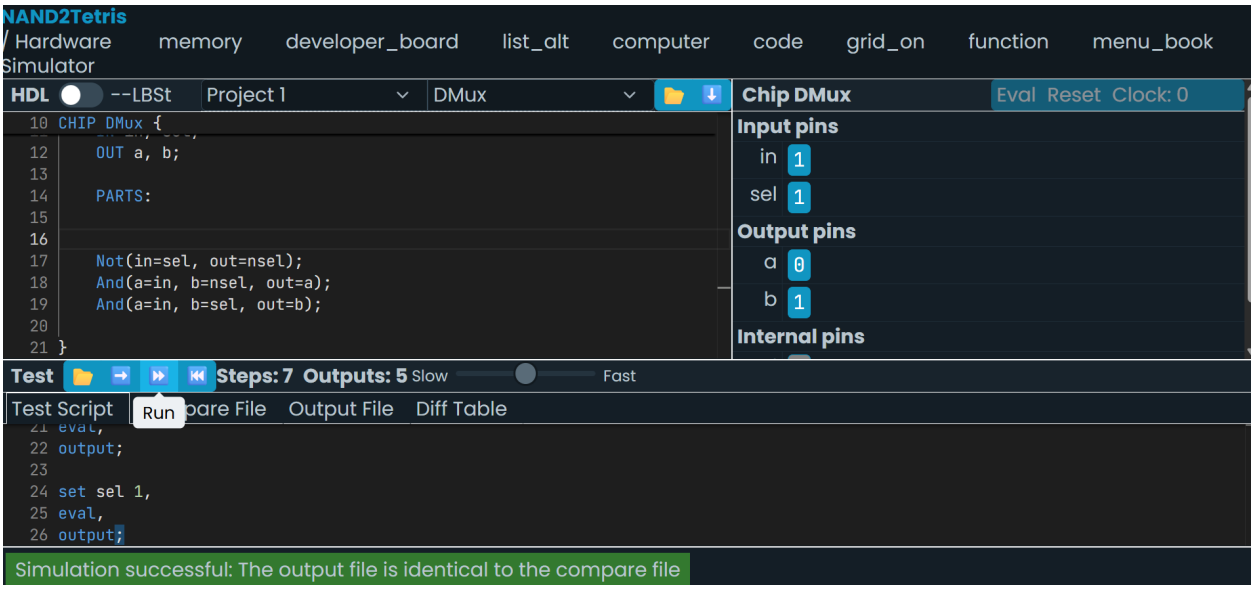
## Chip: Xor


NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
Simulator
HDL
--LBSt
Project 1
Xor
5 /**
9 CHIP Xor {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a, b=b, out=nand1);
15 Nand(a=a, b=nand1, out=nand2);
16 Nand(a=b, b=nand1, out=nand3);
17 Nand(a=nand2, b=nand3, out=out)
18 }
Test
Steps: 7 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Xor.tst
5
6 load Xor.hdl,
7 compare-to Xor.cmp,

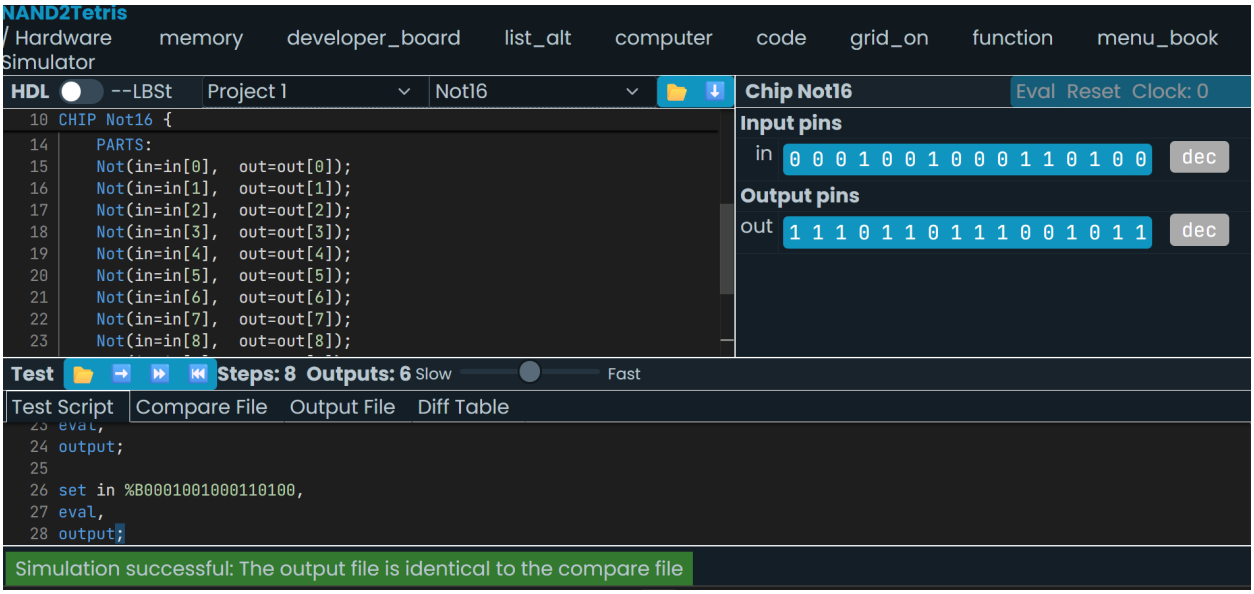
# Chip: Mux


NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
Mux
Chip Mux
Eval
Reset
Clock: 0
5 /**
9 CHIP Mux {
10 IN a, b, sel;
11 OUT out;
12
13 PARTS:
14 Not(in=sel, out=nsel);
15 And(a=a, b=nsel, out=anda);
16 And(a=b, b=sel, out=andb);
17 Or(a=anda, b=andb, out=out);
18 }
Input pins
a 1
b 0
sel 1
Output pins
out 0
Internal pins
Test
Steps: 11
Outputs: 9
Slow
Fast
Test Script
Compare File
Output File
Diff Table
44 output;
46 set sel 1,
47 eval,
48 output;
Simulation successful: The output file is identical to the compare file

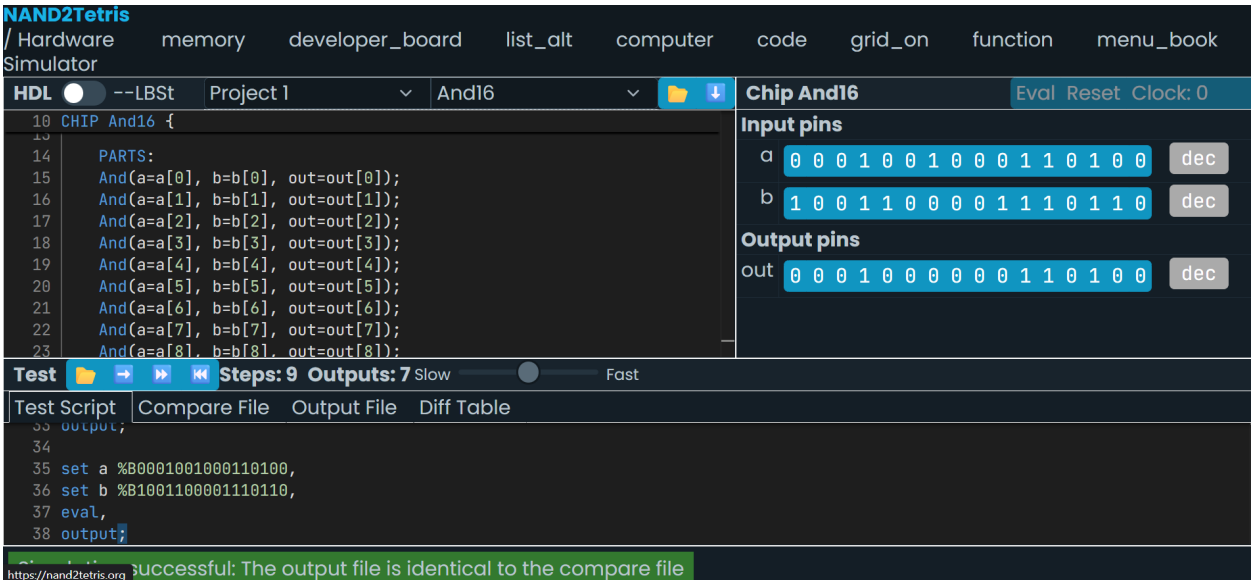
# Chip: DMux


NAND2Tetris
Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
DMux
Chip DMux
Eval Reset Clock: 0
10 CHIP DMux {
12 OUT a, b;
14 PARTS:
17 Not(in=sel, out=nsel);
18 And(a=in, b=nsel, out=a);
19 And(a=in, b=sel, out=b);
21 }
Input pins
in 1
sel 1
Output pins
a 0
b 1
Internal pins
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Run
Output File
Diff Table
22 output;
24 set sel 1,
25 eval,
26 output;
Simulation successful: The output file is identical to the compare file

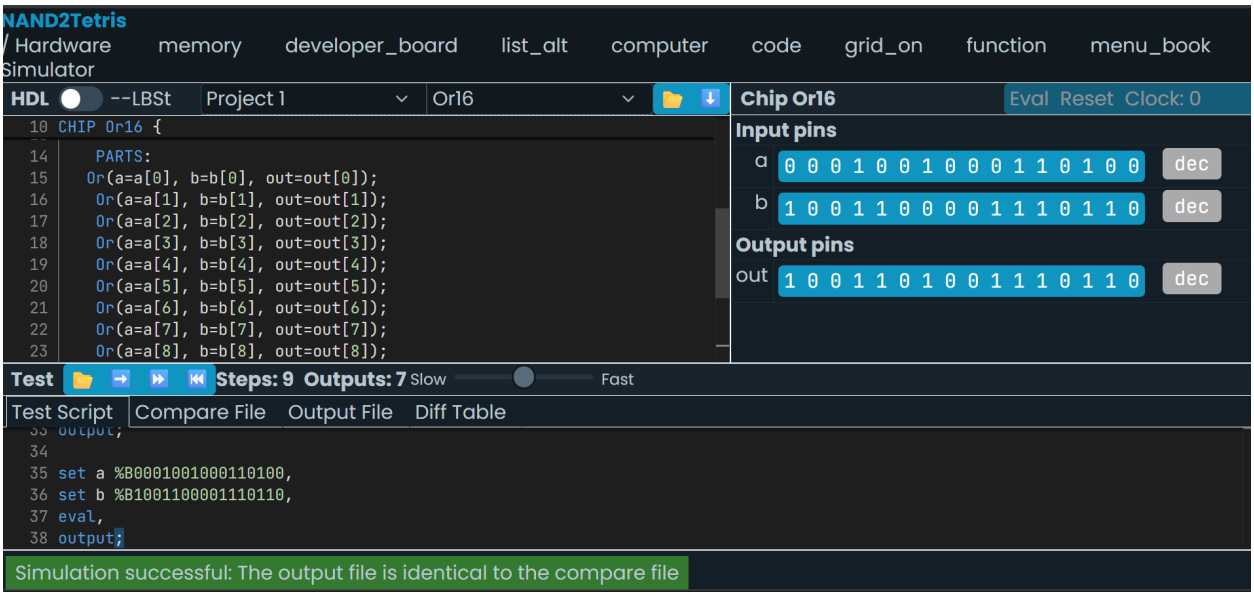
# Chip: Not16


NAND2Tetris
Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
Not16
Chip Not16
Eval Reset Clock: 0
10 CHIP Not16 {
14 PARTS:
15 Not(in=in[0], out=out[0]);
16 Not(in=in[1], out=out[1]);
17 Not(in=in[2], out=out[2]);
18 Not(in=in[3], out=out[3]);
19 Not(in=in[4], out=out[4]);
20 Not(in=in[5], out=out[5]);
21 Not(in=in[6], out=out[6]);
22 Not(in=in[7], out=out[7]);
23 Not(in=in[8], out=out[8]);
Input pins
in 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
Output pins
out 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1
dec
Test
Steps: 8 Outputs: 6
Slow
Fast
Test Script
Compare File
Output File
Diff Table
24 output;
26 set in %B0001001000110100,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

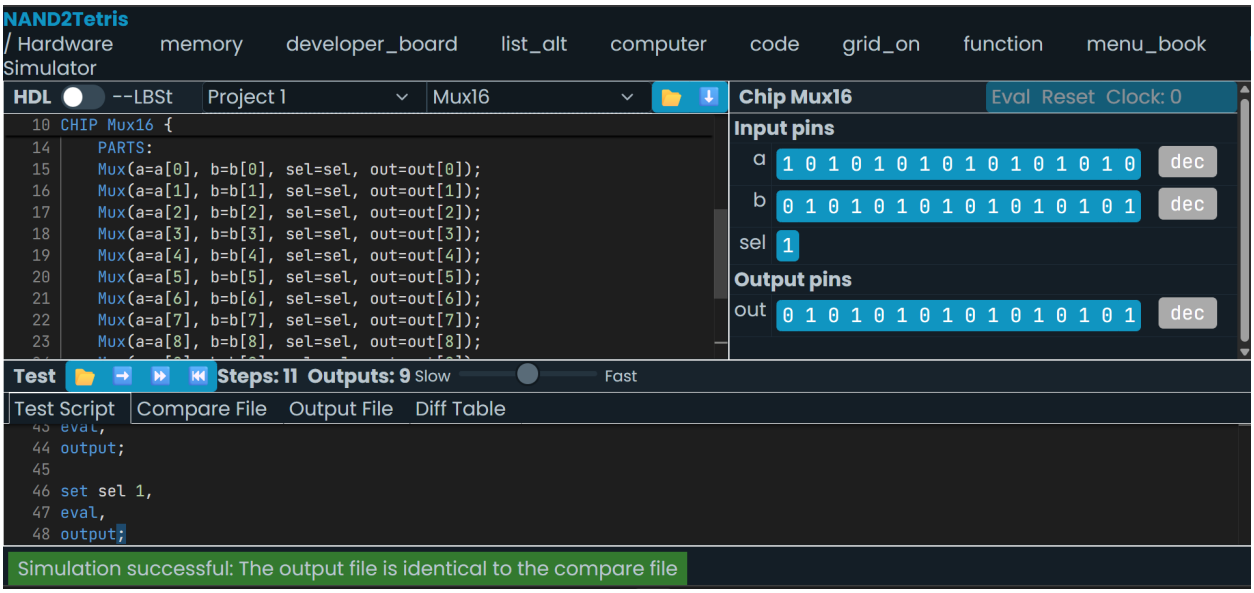
# Chip: And16


NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
And16
10 CHIP And16 {
14 PARTS:
15 And(a=a[0], b=b[0], out=out[0]);
16 And(a=a[1], b=b[1], out=out[1]);
17 And(a=a[2], b=b[2], out=out[2]);
18 And(a=a[3], b=b[3], out=out[3]);
19 And(a=a[4], b=b[4], out=out[4]);
20 And(a=a[5], b=b[5], out=out[5]);
21 And(a=a[6], b=b[6], out=out[6]);
22 And(a=a[7], b=b[7], out=out[7]);
23 And(a=a[8], b=b[8], out=out[8]);
Chip And16
Eval
Reset
Clock: 0
Input pins
a
0 0 0 1 0 0 1 0 0 0 0 1 1 0 1 0 0
dec
b
1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out
0 0 0 1 0 0 0 0 0 0 1 1 0 1 0 0
dec
Test
Steps: 9
Outputs: 7
Slow
Fast
Test Script
Compare File
Output File
Diff Table
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
successful: The output file is identical to the compare file
https://nand2tetris.org

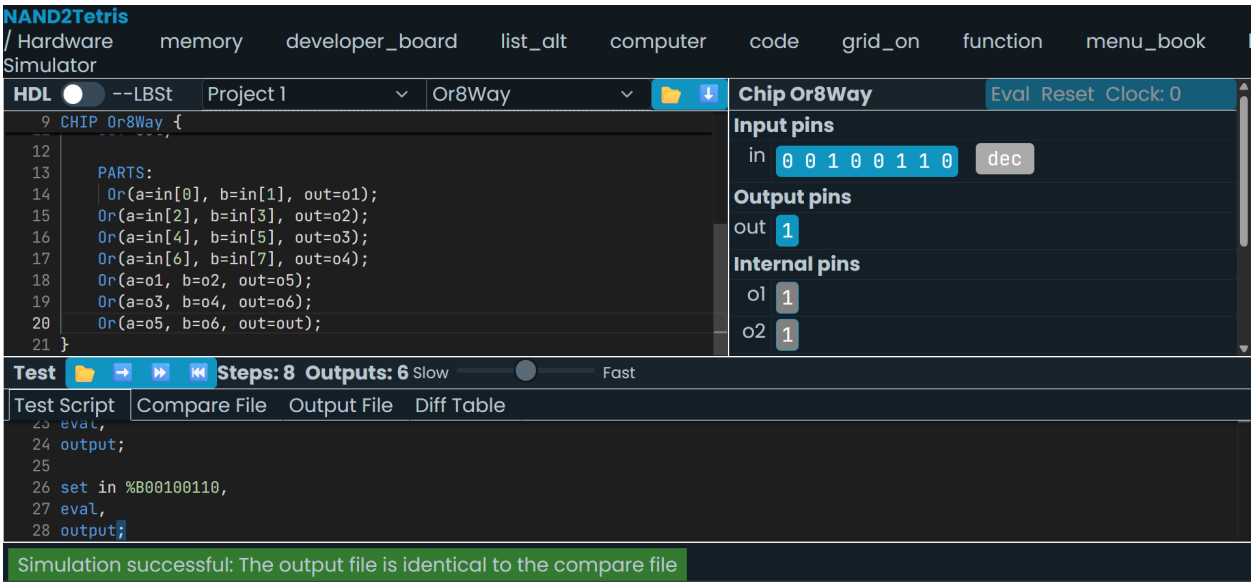
# Chip: Or16


NAND2Tetris
Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
Or16
Chip Or16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 0 1 1 0 1 0 0
dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out 1 0 0 1 1 0 1 0 0 1 1 1 0 1 1 0
dec
10 CHIP Or16 {
14 PARTS:
15 Or(a=a[0], b=b[0], out=out[0]);
16 Or(a=a[1], b=b[1], out=out[1]);
17 Or(a=a[2], b=b[2], out=out[2]);
18 Or(a=a[3], b=b[3], out=out[3]);
19 Or(a=a[4], b=b[4], out=out[4]);
20 Or(a=a[5], b=b[5], out=out[5]);
21 Or(a=a[6], b=b[6], out=out[6]);
22 Or(a=a[7], b=b[7], out=out[7]);
23 Or(a=a[8], b=b[8], out=out[8]);
Test
Steps: 9 Outputs: 7
Slow
Fast
Test Script
Compare File
Output File
Diff Table
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
Simulation successful: The output file is identical to the compare file

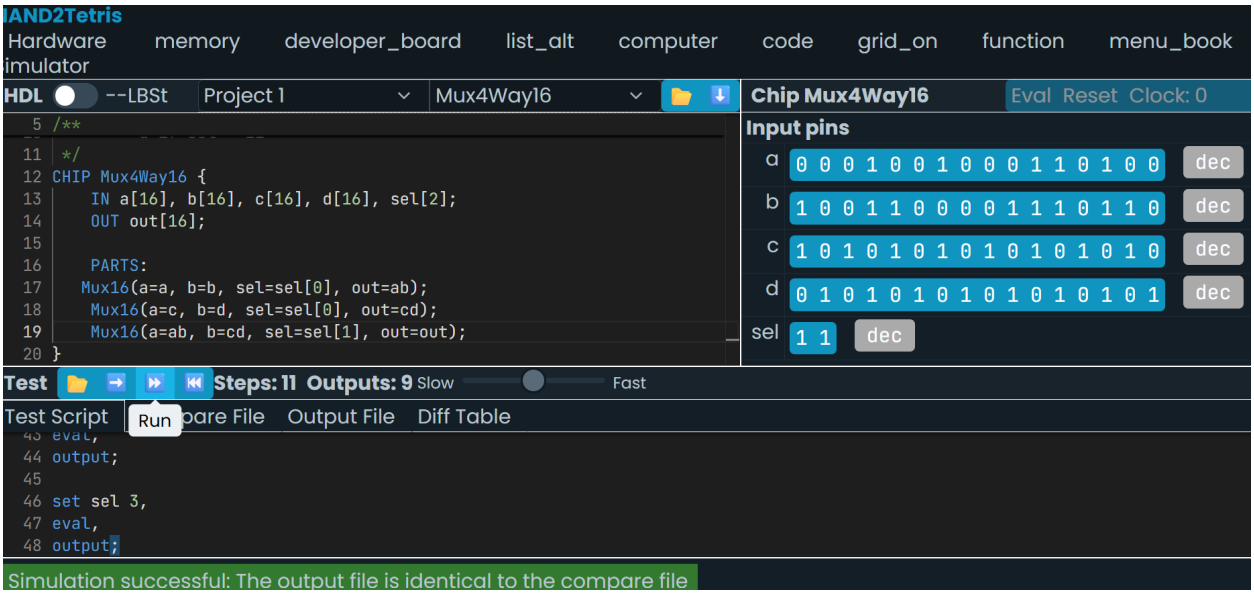
# Chip: Mux16


NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 1
Mux16
10 CHIP Mux16 {
14 PARTS:
15 Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
16 Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
17 Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
18 Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
19 Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
20 Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
21 Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
22 Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
23 Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
Chip Mux16
Eval Reset Clock: 0
Input pins
a 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
dec
b 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
sel 1
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Test
Steps: 11 Outputs: 9
Slow
Fast
Test Script
Compare File
Output File
Diff Table
44 output;
46 set sel 1,
47 eval,
48 output;
Simulation successful: The output file is identical to the compare file

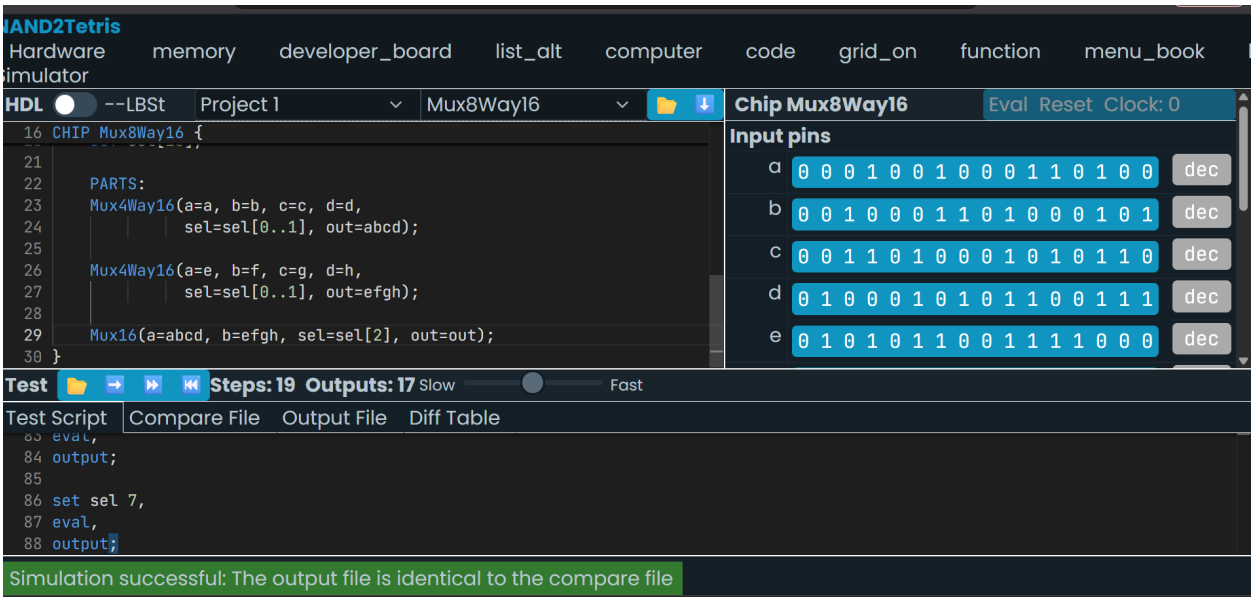
# Chip: Or8Way


NAND2Tetris
/ Hardware memory developer_board list_alt computer code grid_on function menu_book
Simulator
HDL --LBSt Project 1 Or8Way
CHIP Or8Way {
PARTS:
Or(a=in[0], b=in[1], out=o1);
Or(a=in[2], b=in[3], out=o2);
Or(a=in[4], b=in[5], out=o3);
Or(a=in[6], b=in[7], out=o4);
Or(a=o1, b=o2, out=o5);
Or(a=o3, b=o4, out=o6);
Or(a=o5, b=o6, out=out);
}
Chip Or8Way Eval Reset Clock: 0
Input pins
in 0 0 1 0 0 1 1 0 dec
Output pins
out 1
Internal pins
o1 1
o2 1
Test Steps: 8 Outputs: 6 Slow Fast
Test Script Compare File Output File Diff Table
output;
set in %B00100110,
eval,
output;
Simulation successful: The output file is identical to the compare file

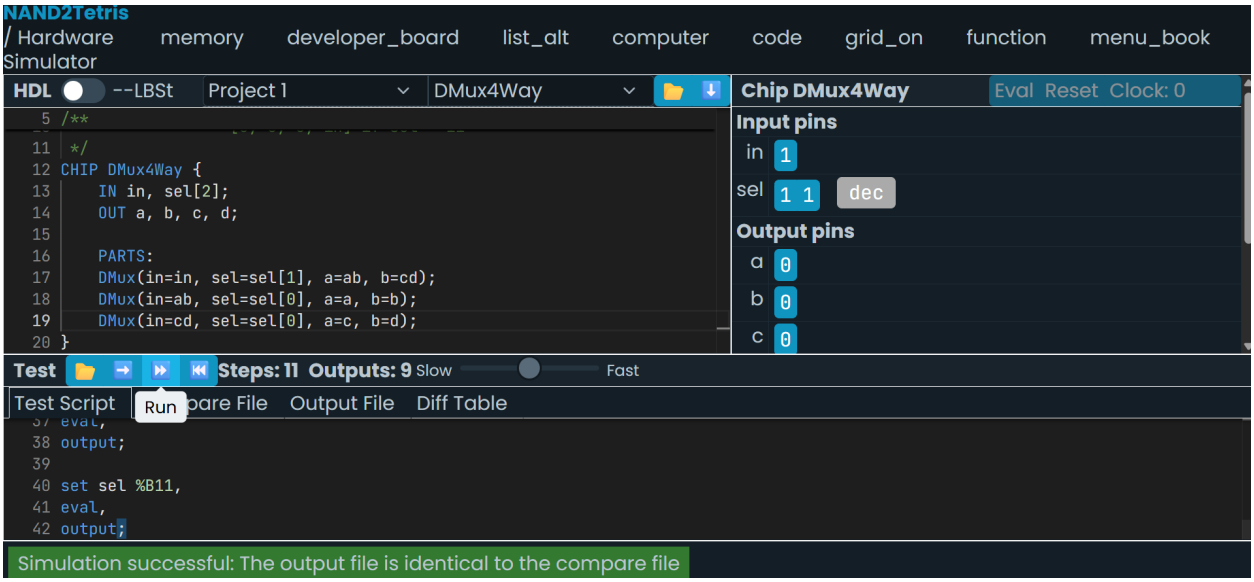
# Chip: Mux4Way16



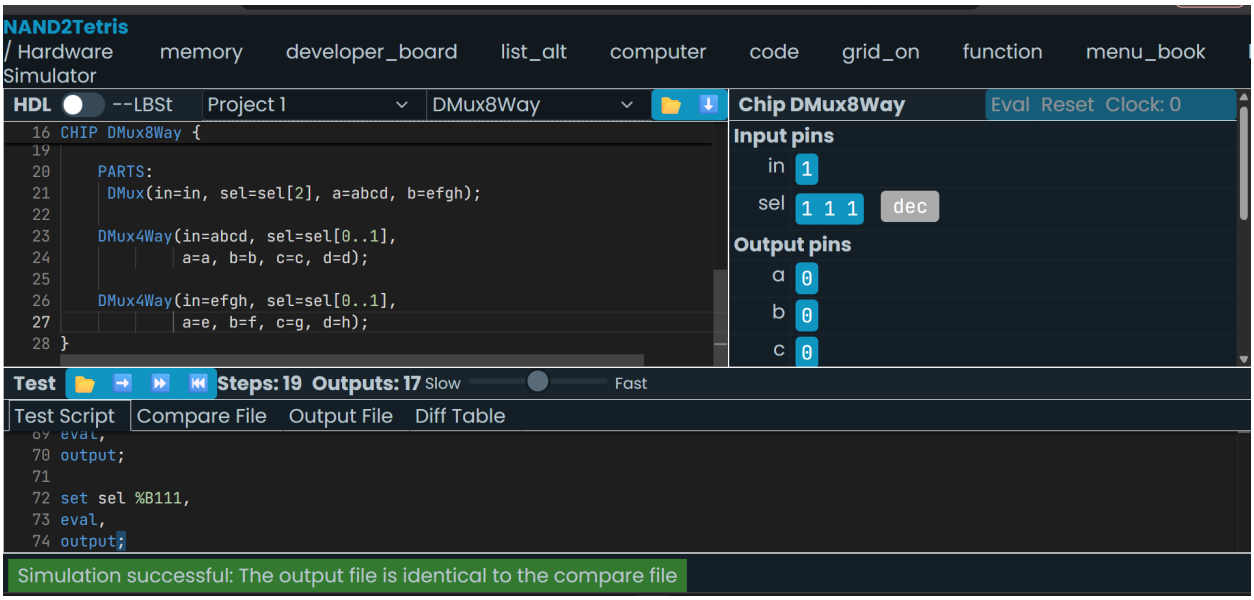
# Chip: Mux8Way16



# Chip: DMux4Way

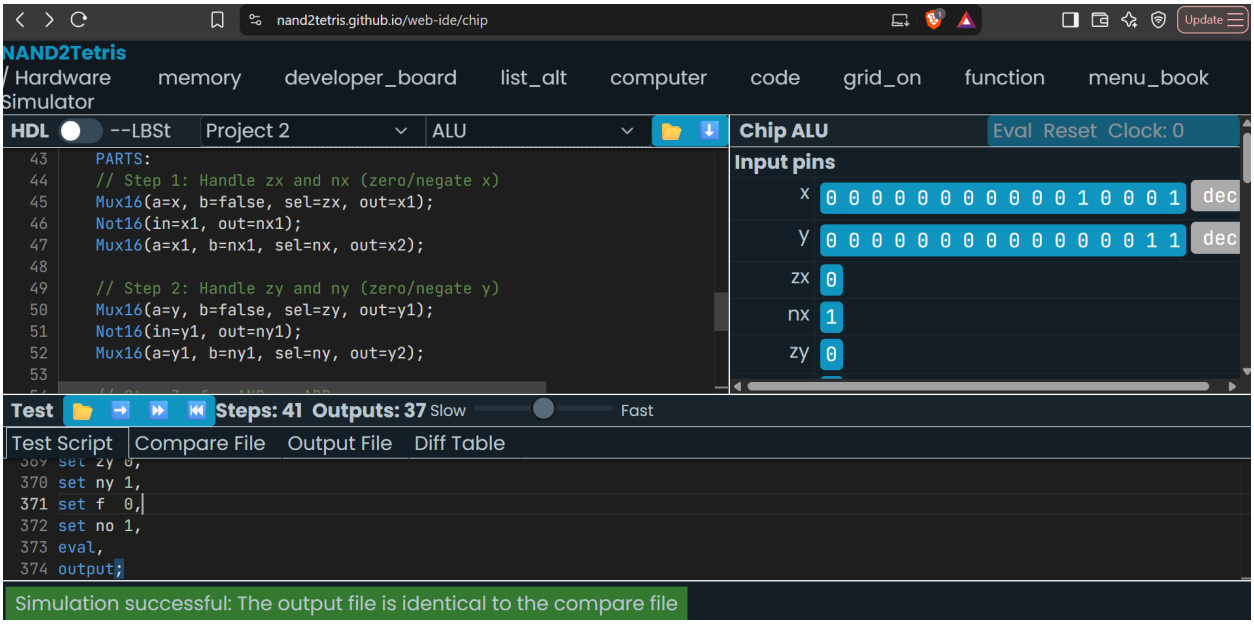


# Chip: DMux8Way

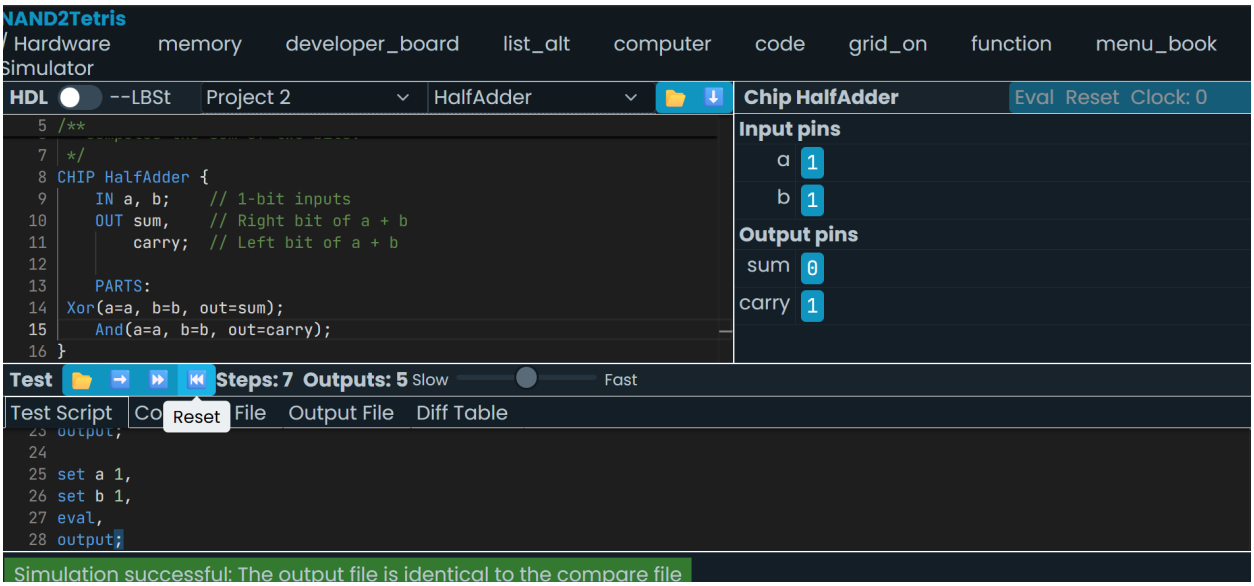


# Project 2

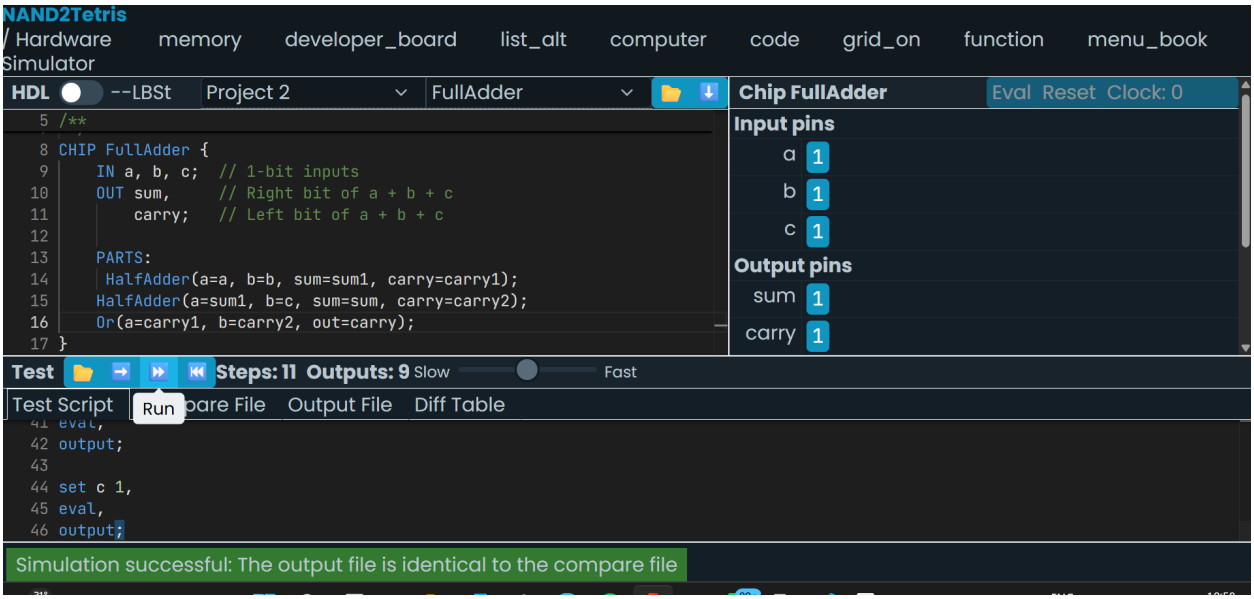
## Chip: ALU



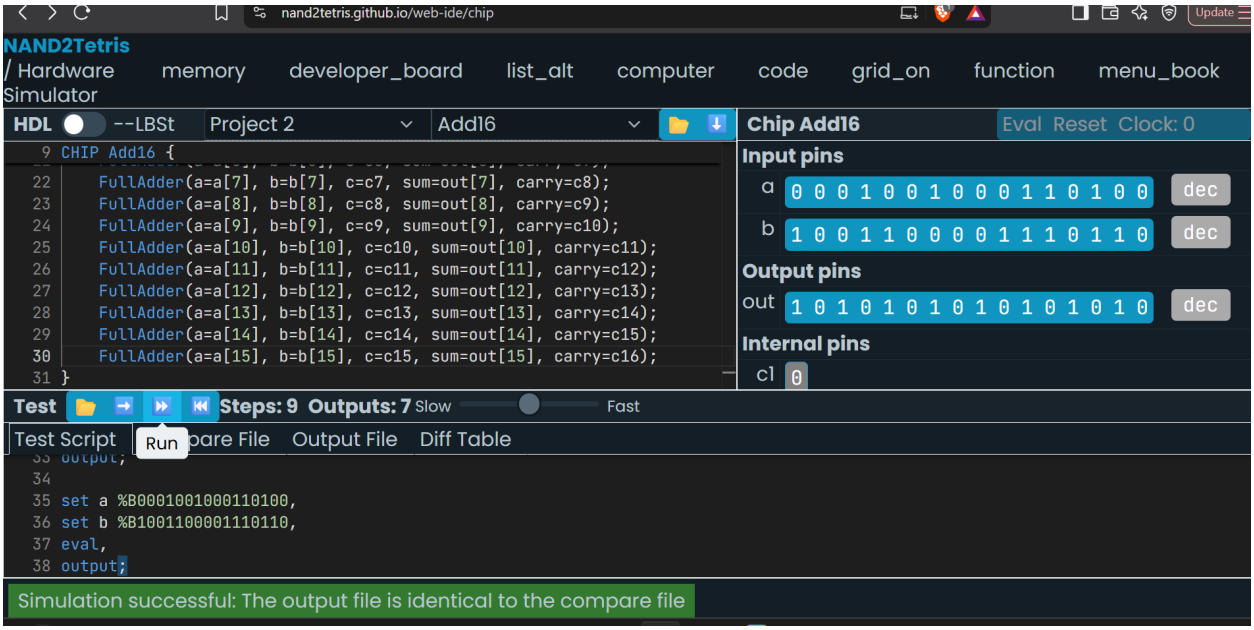
# Chip: HalfAdder



# Chip: FullAdder



# Chip: Add16


nand2tetris.github.io/web-ide/chip
NAND2Tetris
/ Hardware
memory
developer_board
list_alt
computer
code
grid_on
function
menu_book
Simulator
HDL
--LBSt
Project 2
Add16
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 0 1 1 0 1 0 0
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
Internal pins
c1 0
CHIP Add16 {
FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=c12);
FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=c13);
FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=c14);
FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=c15);
FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=c16);
}
Test
Steps: 9 Outputs: 7
Slow
Fast
Run
Test Script
Output File
Diff Table
set a %B0001001000110100,
set b %B1001100001110110,
eval,
output;
Simulation successful: The output file is identical to the compare file